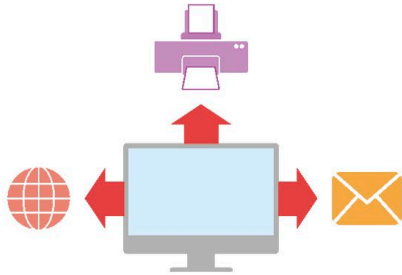


Types of operating system

As computers have progressed and developed, so have operating systems. Within the broad family of operating systems, there are generally four types. These are categorized based on the number of users or applications they can support, and the type of computer they can control.



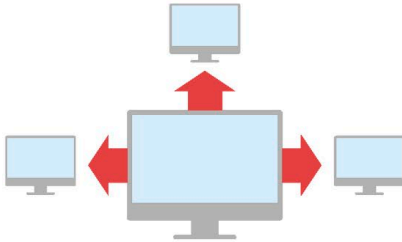
△ Single and multi-tasking

A single-tasking operating system runs only one program at a time, while a multi-tasking OS allows many programs to run simultaneously.



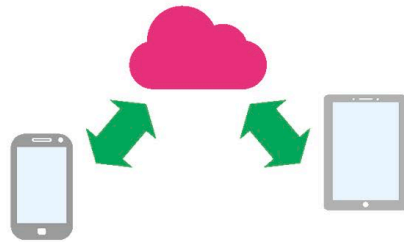
△ Single-user and multi-user

A single-user OS allows only one user to access the computer system at any given time. A multi-user system allows access to many users at a time.



△ Distributed

This OS allows distributed applications to run on several machines that are connected to each other with a high-quality network.



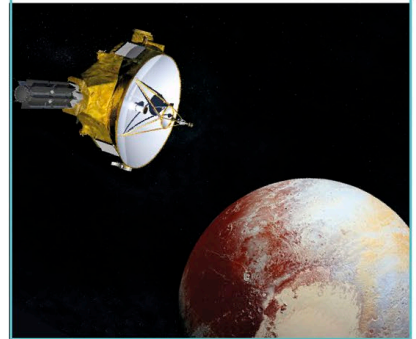
△ Templated

Common in cloud computing, this refers to running multiple virtual machines off a guest OS created on a single computer.

REAL WORLD

RTOS

Real-time operating systems (RTOS) are multi-tasking systems that are used for real-time applications. They are designed for an environment where a large number of events must be processed in a very short time, usually tenths of a second. The US space agency NASA has used RTOS on many missions, including the New Horizons probe to Pluto, launched in 2006. RTOS are cost-effective and do not require a lot of physical hardware in order to run, which is crucial when sending a probe far into space.



Utilities

The operating system uses applications called utilities, which allow users to manage their computer, its devices, and its programs. There are many different utility programs, and they vary across operating systems. Users can access these via a special menu or control panel.



User accounts
and security



Anti-virus
software



Deleting documents,
files, or programs



Software updates



System clean-up



Encryption /
decryption



Word
processing